

# PAPER\_704: Horsehead Nebula (Barnard 33): Infrared Erosion UQFF Analysis

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## Abstract

The Horsehead Nebula (Barnard 33) is a dark molecular cloud in Orion at 1500 ly, photographically famous as a silhouette against the emission nebula IC 434. It is slowly being photoevaporated by ultraviolet radiation from Sigma Orionis ( $L \approx 10^5 L_\odot$ ). The UQFF MUGE is derived for this system incorporating radiation pressure erosion, gravitational self-collapse, and EM aether coupling.

## 1. System Parameters

| Parameter      | Value                        | Description                    |
|----------------|------------------------------|--------------------------------|
| $M_{neb}$      | $120M_\odot$                 | Total mass (100 gas + 20 YSOs) |
| $r_{neb}$      | $1.182 \times 10^{16}$ m     | Half-span (1.25 ly)            |
| $L_{Sigma}$    | $10^5 L_\odot$               | Sigma Orionis luminosity       |
| $\rho_{neb}$   | $10^{-21}$ kg/m <sup>3</sup> | Nebula density                 |
| $\tau_{erode}$ | $1.578 \times 10^{14}$ s     | Erosion timescale (5 Myr)      |
| $E_0$          | 0.2                          | Fractional erosion rate        |

## 2. Master UQFF Gravity Equation

$$g_{HH}(t) = \frac{GM}{r^2}(1 + H_z t)(1 - E(t))(1 + f_{TRZ}) + P_{rad} + q(v \times B) \left(1 + \frac{\rho_{UA}}{\rho_{SCm}}\right) \times 10^{-12}$$

### 2.1 Erosion Term

$$E(t) = E_0 (1 - e^{-t/\tau_{erode}})$$

At  $t = 1$  Myr:  $E \approx 0.039$ , so  $(1 - E) \approx 0.961$ .

### 2.2 Radiation Pressure Acceleration

$$P_{rad} = \frac{L_{star}}{4\pi r_{neb}^2 c} \cdot \frac{\rho_{neb}}{m_H} \approx 4.35 \times 10^{-5} \text{ m/s}^2$$

This represents the net UV ram pressure eroding the nebula pillar.

## 3. Numerical Result

$$g_{HH}(t = 1 \text{ Myr}) \approx 1.097 \times 10^{-3} \text{ m/s}^2$$

## 4. Infrared Observation

Hubble Space Telescope WFC3 IR imaging (2013) reveals the nebula in the NIR, with proto-stellar objects embedded in the pillar tip. The UQFF framework predicts the gravitational collapse rate consistent with the 100 kyr protostellar formation timescale.

## References

- Reipurth & Bally (2001), ARA&A, 39, 403
- Pound (1998), Astrophys. J., 493, L113
- UQFF Framework, Star-Magic Session 175

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